

UNIVERSITY OF TECHNOLOGY, JAMAICA
SCHOOL OF COMPUTING & INFORMATION TECHNOLOGY
Programming 2: Tutorial/Practical Worksheet #1

Tutorial Questions

Part A

Discuss the following terms

1. Definite loops
2. Indefinite loops
3. Counter controlling variable
4. Sentinel Value
5. Continue
6. Break

Part B

For each scenario, identify the error, logical flaw, or inefficiency; state the correct loop type, and provide a brief explanation.

1. A student uses a definite loop to repeatedly ask a user to enter a password until it is correct.
2. A program processes exam scores for exactly 20 students using an indefinite loop.
3. A student uses an indefinite loop to print numbers from 1 to 10.
4. A program is required to accept monthly rainfall data for 12 months. The student stops input when the user enters -1.
5. A definite loop runs 30 times, but the program should stop early if the temperature becomes unsafe.
6. A student uses a definite loop to process customer orders, but the number of customers is unknown.
7. A program uses an indefinite loop to read student marks. However, the maximum number of students allowed is 25, and the loop does not enforce this limit.
8. A program processes 10 employees, but the loop counter starts at 0 and ends at 10.
9. A student selects an indefinite loop because *“all user input should use while loops”*
10. A program uses a definite loop with 50 iterations to process employee data. Some employees leave early, but the loop still runs all 50 iterations.

Part C

1. Observe the following pseudocode and find which statement will never get executed?

```
t = 1                                #statement 1
WHILE t <= 15 DO                     #statement 2
    IF t > 13 THEN                    #statement 3
        PRINT "Binary"               #statement 4
    ELSE                              #statement 5
        PRINT "Octal"                 #statement 6
    ENDIF                             #statement 7
    t = t + 3                         #statement 8
ENDWHILE                             #statement 9
```

- a. Statement 3
 - b. Statement 4
 - c. Statement 6
 - d. Statement 8
2. Write the value of sum after execution of the following loop?

```
i = 1
sum = 0
WHILE i < 10 DO
    sum = sum + 1
    i = i + 2
ENDWHILE
```

- a. 24
 - b. 25
 - c. 26
 - d. 27
3. Rewrite the pseudocode from Questions 1 and 2 using:
- a. a *for* loop
 - b. a *do-while* loop

4. Write a pseudocode that will keep asking for a number until the user enter a negative number. At the end, print the sum of all positive number entered.
5. Identify and correct all syntax, logical, and loop errors to ensure the program works as intended.

```
a. #include <stdio.h>

int main()
{
    int number;
    int sum = 0;
    int count;

    while (number != 0)
    {
        printf("Enter a positive number (0 to stop): ");
        scanf("%d", &number);

        if (number < 0);
        {
            printf("Invalid number\n");
        }
        else
        {
            sum = sum + number;
            count++;
        }
    }
    printf("Sum: %d\n", sum);
    printf("Numbers entered: %d\n", count);
    return 0;
}
```

```

b. #include <stdio.h>

int main()
{
    int count = 0;
    int total;
    float average;
    int mark;

    printf("Enter student marks (-1 to stop):\n");

    for (count = 1; count <= 10; count++);
    {
        printf("Enter mark for student %d: ", count);
        scanf("%d", mark);

        if (mark = -1)
            break;

        if (mark < 0 && mark > 100)
        {
            printf("Invalid mark entered\n");
        }
        else
        {
            total = total + mark;
        }
    }

    average = total / count;

    printf("Total Marks: %d\n", total);
    printf("Average Mark: %.2f\n", average);
    printf("Number of Students: %d\n", count);

    return;
}

```

Challenge

The Portland Rafting Company would like a program to manage its daily operations of providing customers with personalized bamboo raft trips on the river lasting 1, 2 or 3 hours. Three packages are offered, which provide extras along with the trip on the river, the:

- Scenic Tour (base cost - US\$119), which includes stopping for drinks or snacks

- Relaxation Tour (base cost - US\$139), which includes stopping at popular landmarks or for photos
- Joyous Fun Tour (base cost - US\$149), which allows for swimming.

The number of hours determines how many different landmarks (in the case of the Relaxation Tour), the length of the drink/snack stop (in the case of the Scenic Tour) or the length of time spent swimming (in the case of the Joyous Fun Tour). In addition to the package cost, each hour spent on the trip attracts a cost of US\$20.

Write the **pseudocode** for a program that will accept a series of records. Each record contains a customer ID, customer name, the number of hours that the customer would like to spend on the trip and the type of package selected. Based on the package type and the number of hours, the program should determine the cost of the trip for EACH customer.

Output for the program consists of the customer ID, customer name and the cost of the trip for each customer. It should also output the total number of customers processed for the day and the overall total cost of all trips made by customers on any particular day. The program should use -1 for the customer ID to signal that no more records are to be processed. **Assume valid input.**

In the Laboratory

Complete any unfinished questions from the tutorial section. Implement and execute the following programs using both **C** and **Python**:

1. A small convenience store records its sales for 7 days. Each day's sales amount is entered in dollars. Write a program to input the sales amount for each of the 7 days. Calculate and display the total weekly sales.
2. A system requires the user to enter a password. The correct password is "secure123". Write a program that will repeatedly prompt the user for the password. It should only stop when the correct password is entered. Display a success message once access is granted.
3. A company employs 40 workers. Each worker's weekly salary is entered. Develop a solution that will input each employee's salary. Calculate the total payroll cost and count how many employees earn more than \$80,000.00.

4. A fast-food restaurant allows customers to place multiple orders. Each item costs \$500. Write a program that will repeatedly ask if the customer wants to order another item (Y/N). Calculate and display the total cost when ordering is complete.
5. A teacher enters student marks one at a time. Entry stops when -1 is entered. Valid marks range from 0 to 100. Write a program to
 - a. accept marks until -1 is entered.
 - b. Count how many valid marks were entered.
 - c. Calculate and display the average mark
6. A supermarket allows up to 10 customers to check out. Each customer may buy an unknown number of items. Each item price is entered individually. Develop a solution that will:
 - a. process up to 10 customers
 - b. allow each customer to enter items until they choose to stop
 - c. calculate and display each customer's bill